MIL-C-44317(GL)
21 January 1988

MILITARY SPECIFICATION

CLOTH, LAMINATED, NYLON, WATER REPELLENT TREATED:

FOR INDIVIDUAL MULTIPURPOSE SHELTER (IMPS)

This specification is approved for use by the U.S. Army Natick, Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a water repellent treated nylon laminated cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

P-D-245	-	Detergent, Laundry and Hand Dishwashing (Granular)
PPP-P-1136	-	Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. NATAICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1516B - Woodland Pattern - 60 Inches

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ANERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2582 - Puncture Propagation Tear Resistance of Plastic Film and Thin Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The printed laminated cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.8).

3.3 Materials.

3.3.1 Base cloths.

3.3.1.1 Plain weave. The plain weave cloth shall be texturized nylon, 2.7 ± 0.2 ounce per square yard, Woodland Camouflage Pattern. The cloth shall be woven with a minimum of 101 ends per inch and 63 picks per inch. The cloth shall have a minimum breaking strength of 135 pounds in the warp and 100 pounds in the filling.

3.3.1.2 Tricot knit. The tricot knit cloth shall be nylon, 1.6 ± 0.3 ounces per square yard, Olive Green 106. The cloth shall be knit on 2 bars using 40 denier multifilament and no less than 45 denier monofilament yarn. The fabric shall have a minimum bursting strength of 100 pounds.

3.3.2 Thermoplastic film. The thermoplastic film shall be polyurethane, weighing 0.9 ± 0.2 ounces per square yard.

3.3.3 Adhesive. The adhesive for laminating the layers together shall be such that the laminated cloth will meet the requirements of this document. The adhesive shall not be a health hazard when used as intended.

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3.4 Laminated cloth. The plain woven base cloth shall be laminated to one side of the thermoplastic film and the tricot knit fabric shall be laminated to the other side of the thermoplastic film. The face side of the laminated cloth shall be the side with the Woodland Camouflage print plain weave fabric. An approved Quarpel type water repellent finish shall be applied to the cloth. The laminated cloth shall conform to all the requirements of table I when tested as specified in 4.4.3.

TABLE I. Laminated cloth requirements

Characteristics	Minimum	Maximum
Weight, oz/sq.yd.	5.8	6.8
Breaking strength, lbs.		
Warp	150	-
Filling	130	-
Tearing resistance, kilograms-force(kgf)		
Warp	3.9	-
Filling	3.9	-
Stiffness (warp only), cm at 70°F	-	12.0
Hydrostatic resistance, psi	95	-
Water permeability, initial	No leakage	-

3.4.1 Color. The color of the face side of the laminated cloth shall be Woodland Camouflage Pattern. The color of the back side of the laminated cloth shall be OG 106. Both sides of the fabric shall match the respective sides of the standard sample.

3.4.2 Matching. The color and appearance of the camouflage printed cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.4.3 Colorfastness. The camouflage printed, laminated cloth shall meet the following colorfastness requirements when tested in accordance with 4.4 for the characteristics listed below:

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Characteristic	Colorfastness requirements
Fastness to light (after 40 hours) and fastness to laundering (after 3 cycles)	Equal to or better than the standard sample, or equal to or better than "good".
Fastness to perspiration	Equal to or better than the standard sample. or equal to or better than "good". 1/
Fastness to crocking	Equal to or better than the standard sample, or not less than the AATCC Chromatic Scale Rating of 3.5. 2/

1/ Except the fastness of Dark Green 355 and Brown 356 shall be not less than "fair" when tested for fastness to alkaline perspiration.

2/ Except Black 357 shall show an AATCC Chromatic Transference Scale rating of not less than 1.0.

3.4.4 Pattern execution. The pattern shall reproduce the standard sample in respect to design, colors, and registration of the respective areas. The pattern repeat of the printed and laminated cloth shall be 27.25 + 1.25, - 2.50 inches in the warp direction. Each pattern area shall show solid coverage. Skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution on design, the pattern of the base cloth shall match the standard Woodland Camouflage Pattern Drawing 2-1-1516B (see 2.1.1).

3.4.5 Spectral reflectance. The camouflage printed end water repellent treated side of the finished cloth shall conform to the spectral reflectance requirements specified in table II when tested as specified in 4.5.1. The spectral reflectance values, for Black 357, in the wavelengths region 700 to 860 nm shall be 10 percent or less relative to barium sulfate, the preferred white reflectance standard, or any other white reference standard specified in 4.5.1.

TABLE II. Spectral reflectance requirements

Wavelength nanometers	Reflectance values (percent) Light Green 354 Min.	Light Green 354 Max.	Dark Green 355 Brown 356 Min.	Dark Green 355 Brown 356 Max.
600	10	20	4	8
620	10	20	4	8
640	9	20	4	8
660	8	20	3	8
680	8	32	3	22
700	14	53	9	41

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TABLE II. Spectral reflectance requirements

Wavelength nanometers	Reflectance values (percent) Light Green 354 Min.	Max.	Dark Green 355 Brown 356 Min.	Max.
720	35	67	25	60
740	55	73	50	73
760	60	78	55	78
780	60	80	57	80
800	60	81	59	81
820	60	82	60	82
840	60	82	60	82
860	60	82	60	82

3.4.6 Water repellency. The laminated cloth shall be given an approved Quarpel type water repellent treatment on the face side of the cloth and shall conform to the water repellency requirements. The use of materials other than approved water repellents are prohibited (see 6.4).

3.4.7 Spray ratings. The finished cloth shall have a spray rating equal to or better than 100, 90, 90 initially and 70, 70, 70 after three launderings when tested as specified in 4.4.3.

3.4.8 Resistance to organic liquids. The finished cloth shall show no wetting by n-tetradecane, initially and after 3 launderings when tested as specified in 4.4.3.

3.4.9 Resistance to delamination. Delamination between the cloth and film after 5 launderings shall not exceed 0.25 linear inches when tested as specified in 4.5.3.

3.4.10 Dimensional stability. The shrinkage or elongation in the warp direction of the finished cloth shall be not greater than 4.5 percent for the individual sample unit and not greater than 4.0 percent for the lot average, and the filling direction of the finished cloth, the shrinkage or elongation shall not be greater than 2.5 percent for the individual sample unit and not greater than 2.0 percent for the lot average when tested as specified in 4.4.3.

3.5 Width. The minimum overall width shall be as specified (see 6.2). Selvages shall be trimmed to give straight, uniform, fully laminated edges.

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3.6 Length and put-up. Unless otherwise specified (see 6.2), the laminated cloth shall be furnished in rolls. Each roll shall contain no more than 5 pieces and no single piece shall be less than 10 yards in length. Acceptance of shorter lengths, if permitted, shall be based on the conditions stipulated in the contract or purchase order (see 6.2). The cloth shall be put-up on full width rolls as specified in 5.1.

3.7 Fiber identification. Each roll shall be labeled, and ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship. The laminated cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the acceptable quality levels. Repairs of any kind, of the finished, laminated cloth or of the component materials of the laminated cloth are not permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for defects listed in table III and shall be tested for the characteristics specified in table V. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Thermoplastic film overall weight test. The thermoplastic film shall be tested in accordance with Method 5041 of FED-STD-191 except that each specimen shall be 25 square inches and shall be cut in diagonal fashion from each sample unit. The specimens shall be equally spaced across the full width of the sample unit no closer than 2 inches to the edges of the sample unit. The distance between the top of a specimen and the bottom of the next specimen shall be 3 inches. No two specimens shall overlap in the width or lengthwise direction. The sample unit shall be 1.5 yards full width of the plastic film. The lot shall be rejected if any specimen weighs less than 0.7 or greater than 0.11 ounces per square yard, or if the difference in weight between any two specimens in the sample unit is greater than 0.25 ounces per square yard. The sample size shall be in accordance with the following:

Lot size (yards)	Sampling size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 up to and including 150,000	5

4.4.1.2 Certification. Components listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements.

Component	Requirement paragraph
Plain weave nylon cloth	3.3.1.1
Tricot knit nylon cloth	3.3.1.2

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The cloth shall be examined on both sides for the defects listed in table III. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor combined) defects for the face side of the laminated cloth and 4.0 for

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major defects and 10.0 for total (major and minor combined) defects for the back side of the laminated cloth, but with an AQL of no more than 6.5 total combined major defects for the front and back sides of the laminated cloth. Any defects scored on one side of the laminated cloth shall not be scored on the opposite side. The number of rolls from which the sample yardage is to be selected shall be in accordance with table IV. The sample yardage shall be apportioned equally among the selected rolls.

TABLE III. End item visual defects

Defects	Classification Major	Minor
Any hole, cut, or tear, including edges	X	
Abrasion-resulting in a thin or weak place	X	
Floats or skips - multiple, 1/2 inch or more in either warp or filling direction		X
Blisters, tunnels or delamination of cloths	X	
Crease or wrinkle - resulting in doubling that cannot be corrected by manual pressure, or adhesion of surfaces against each other, or any diagonal distortions of woven (face) fabric surface	X	
Any solid lump, defined as a slub C or 4, or knot which exceeds level C, on the respective Sears Fabric Defect Scales (see 6.7).	X	
Fabric edges rolled, folded, doubled, scalloped or wavy	X	
Any foreign matter 1/		X
Width less than minimum specified	X	
Any odor other than that which is characteristic of the laminating compound or water-repellent finish		X
Any spot, stain, or color off shade, not uniform, mottled, or spotted 2/		X
Any tackiness	X	

1/ Foreign matter shall be defined as waste, fly, or extraneous material that has been knit into fabric or spun into yarn.

2/ Woven (face) fabric side only.

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TABLE IV. Sample size

Lot size in yards	Sample size in rolls
1200 or less 1/	3
1201 up to and including 3200	5
3201 up to and including 10,000	8
10,001 up to and including 35,000	13
35,001 up to and including 150,000	20
150,001 and over	32

1/ If lot contains fewer than three rolls, each roll in the lot shall be examined.

4.4.2.2 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.4.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. If any one component color in any roll of the sample is off shade or does not have the same appearance as the standard sample it shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.2.5 Repairs. During the yard-by-yard examination, each roll in the sample shall be examined for the presence of repairs of any type to the finished, laminated cloth or any of its component materials. The lot shall be unacceptable if nay roll in the sample contains a repair.

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table V shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for all testing shall be 5 continuous yards full width of the finished cloth supplied on a tube to prevent folding. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The maximum lot size shall be 150,000 linear yards. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 up to and including 150,000	5

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method 1/
Overall weight	3.4	5041
Breaking strength	3.4	5100 2/
Tearing resistance	3.4	ASTM D2582 3/
Hydrostatic resistance	3.4	5512 4/
Stiffness (warp only), cm	3.4	5204
Water permeability	3.4	5516 5/
Colorfastness to:		
Light	3.4.3	5660
Laundering (after 3 cycles)	3.4.3	5614 6/
Perspiration	3.4.3	5680
Crocking	3.4.3	5651
Spectral reflectance	3.4.5	4.5.1
Water repellency materials	3.4.6	7/
Spray rating:		
Initial	3.4.7	5526
After 3 launderings	3.4.7	5552 and 5526
Resistance to organic liquid:		
Initial	3.4.8	4.5.2
After 3 launderings	3.4.8	5552 and 4.5.2
Resistance to delamination	3.4.9	4.5.3
Dimensional stability	3.4.10	5552

1/ The face side of the laminated cloth shall be used to orient the test specimens in either the warp or filling direction as required by the applicable test method.

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2/ Measurements shall be read from either dial or recording mechanism and a tensioning clamp shall not be required.

3/ Five warp and five filling specimens shall be tested after conditioning in accordance with section 4 of FED-STD-191. Specimen size shall be 8 inches by 8 inches. Only one tear shall be made on a single specimen. The specimen shall be positioned with the face side toward the probe and with the designated yarns of the face fabric at right angles to the direction of tear. The test shall be conducted using carriage No. 4 and the standard drop height of 508 ± 2 mm. If the tear is not straight on both sides of the laminate, the result shall be considered invalid and another specimen shall be tested.

4/ Camouflage print facing water.

5/ 50 cm hydrostataic head pressure for 5 minutes.

6/ Specimen size shall be 4 grams ± 0.1 grams.

7/ The contractor shall report the water repellents used and certify that no other material has been added.

4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Spectral reflectance test. Reflectance data shall be obtained from 600-860 nm relative to a barium sulfate standard, the preferred white reference standard. Other reference white standards may be used provided they are calibrated to an absolute white; e.g., Halon, magnesium oxide or vitrolite tile. The spectral bandwidth at 860 nm shall be less than 26 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CIE Source A or CIE Source D65. Each shade of the pattern shall be measured as a single layer of laminate backed with six layers of unlaminated outer shell material of the same shade. Readings shall be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal with specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter.

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4.5.2 Resistance to organic liquid test. Place a small specimen of the cloth on smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After 30 seconds examine the specimen under light at an angle. Absence of light reflectance at the cloth drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample unit shall be tested. Evidence of wetting on one or more specimens shall be considered a test failure.

4.5.3 Delamination test. Place 2 ± 0.2 pounds of laminated cloth (one sample unit, 24 inches by full width plus ballast) in a top-loading, commercially available, home-type washer set on the permanent press (10 minutes sudsing) cycle using a high water level (10 inches). The water temperature shall be warm (100 ± 10°F). Place 0.5 ounce of detergent conforming to type II of P-D-245 into the washer. The duration of each laundering cycle shall be 30 ± 5 minutes. After laundering, place sample and ballast in a commercially-available home-type dryer set on the permanent press cycle using medium heat (100 ± 10°F) and dry for approximately 15 minutes. Repeat laundering and drying cycle five times. After the fifth drying cycle, examine both sides for delamination. The sample shall be considered delaminated if the separation between any two plies exceeds 0.25 inch in any direction.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth, shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required in the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The cloth is intended for use in the Individual Multi-purpose Shelter (IMPS).

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. First article (see 3.2, 4.3, 6.8)
c. Width of cloth required (see 3.5).
d. Put-up required, if other than specified (see 3.6).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Quarpel water repellent. The "Quarpel type" water repellent treatment consists of the co-application of an emulsified fluorocarbon and a fluorocarbon extender. Approval of components and combinations is the responsibility of U.S. Army Natick Research, Development, Engineering Center, Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity which are not set forth in this document. Because of the time necessary to conduct full evaluation (approximately 6 months), only those chemical treatments already approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement.

6.5 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the near-infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR and Macbeth 1500 with IR option.

6.6 White standards. Barium sulphate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company.

6.7 Fabric defect scales. Fabric Defect Replica Kits are available from Sears Roebuck and Company, Department 817, (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

6.8 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.9 Subject term (key word) listing.

Cloth, laminated, nylon, water repellent treated
Indivisual multipurpose
Shelter

Custodians

Army - GL

Preparing activity:

Army - GL

Project No. 8305-A221

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44317(GL)	Cloth, Laminated, Nylon, Water Repellent Treated: For Individual Multipurpose Shelter (IMPS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 NOV 84

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